One bike. One connected system.

Breadboard-first proposal, firmware draft and animated diagnostic dashboard.



Pico for the ride. ESP for the records.

Local sensing and lighting, connected trip data, and a handheld trackball remote. Built around your photographed parts, with unknown ratings left explicit.

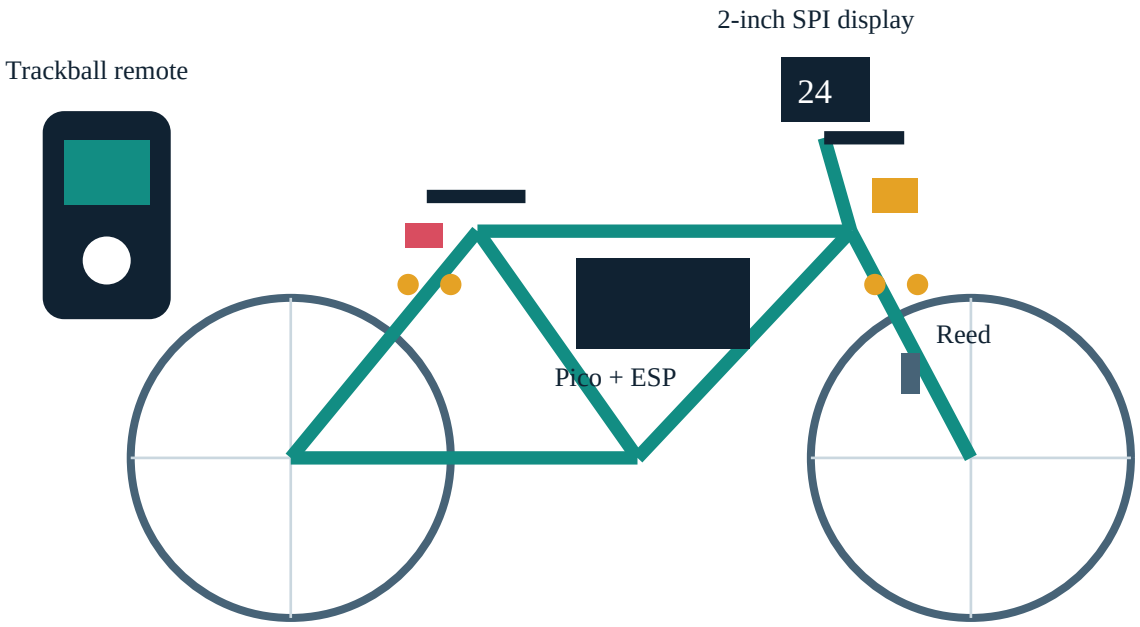
DESIGN STATUS

Runnable software simulation and input-only Pico demo. Hardware drivers, wireless security, GPS logging and parking mechanics still require integration and bench verification.

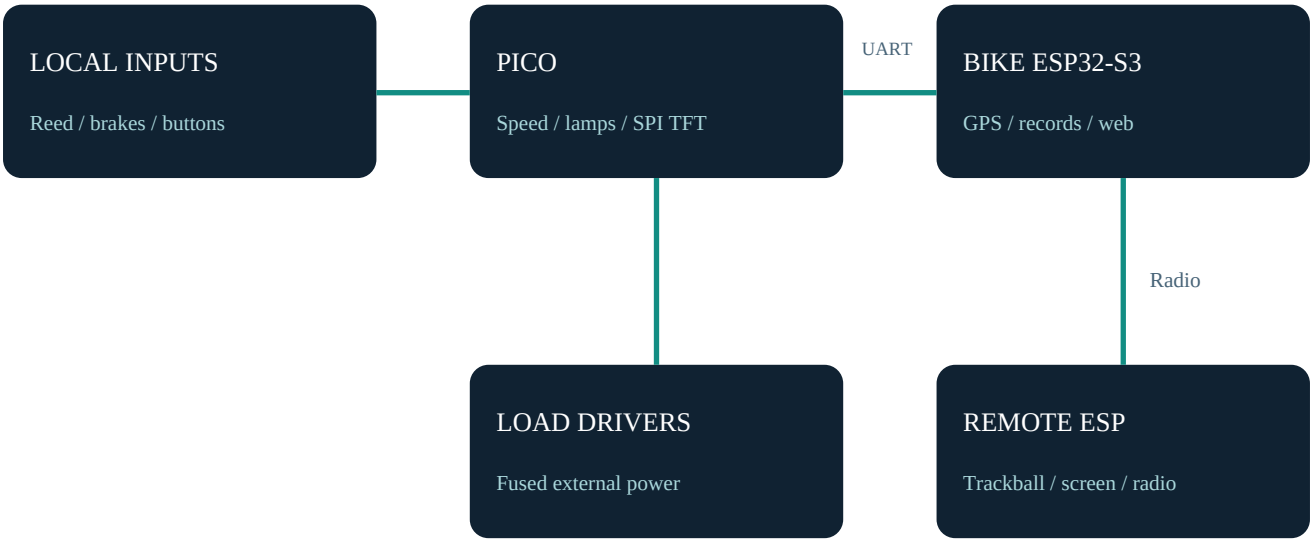
Cover: AI-generated concept, not your actual bicycle or an electrical drawing. Following diagrams are vector artwork.

PHYSICAL & LOGICAL LAYOUT

Equipment around the bike



Handlebar: TFT and buttons. Frame: Pico, ESP and protected power. Fork: reused reed sensor. Front/rear: lamps. Remote: second ESP, trackball and its own display.



Functional allocation only. Not a verified circuit or pinout.

Lock output is unimplemented. Use independent mechanical parking interlocks before considering actuator integration.

1. Scope and evidence

Use a Raspberry Pi Pico for deterministic local functions: reed wheel sensor, 2-inch SPI display, brake input, handlebar buttons and front/rear lamp logic. Use the ESP32-S3 for telemetry, GPS, persistent trip summaries, local web UI and remote communication. Local brake/lighting operation must not wait for Wi-Fi, GPS or the ESP. Keep an independent usable lighting fallback during development.

The supplied photographs show: RP2040 Pico listing; 2.0 TFT SPI GMT020-02 VER1.1 board with BLK/CS/DC/RST/SDA/SCL/VCC/GND; GOOUUU ESP32-S3-CAM-labelled board; trackball breakout with directional/button/RGB pins; NEO-6M on GY-GPS6MV2; E220-900T22D radio modules; salvaged high-power LED/driver and reed sensor; TrustFire EB02 box labelled 8.4 V output and USB 5 V maximum 2100 mA. Labels establish identity clues, not verified electrical performance. Screen controller, exact board variant, lamp ratings and cell protection remain unknown. The uploaded PDF contains generic product safety text, no circuit or pinout.

Purchased mechanical items include brake/shift cables, housings, cable ends, chain tool, cassette/freewheel tool, handlebar extension, jumper wires and an assorted component kit. Inspect suitability: brake housing must be brake-rated; 4 mm versus 5 mm alone does not establish safety or compatibility. Diagnose chain skipping and restore braking before any riding tests. Electronics do not repair mechanical drivetrain or brake faults.

2. Three nodes and communication

Pico: live speed, units, circumference/magnet calibration, display brightness, local brake and indicator timing. Amber front/rear signals on the selected side share one phase (prototype 500 ms on/500 ms off). Brake red overrides rear idle brightness without suppressing turn signals. Main beam and idle lamps are separate logical channels; actual wiring depends on inspecting the light assembly.

Bike ESP32-S3: receives Pico telemetry, reads GPS, stores trip summaries, serves a local dashboard and manages authenticated remote presence. Proposed UART allocation: UART0 Pico, UART1 GPS, UART2 E220. Use native USB for debugging so the three UARTs are not consumed by the debug console. Board USB routing, boot logs on UART0 and GPIO routing must be verified; isolate external equipment during flashing. Camera/PSRAM/flash reservations may constrain the photographed board.

Remote: second ESP, second E220 if using the long-range link, separate screen, trackball, protected battery and charging. Trackball up/down moves between Alarm, Lock, Auto lock and Night mode; centre selects; left/back cancels. Show pending, acknowledged and failed commands distinctly. RGB LED can indicate link state. Confirm the trackball's power and output type before using 3.3 V pull-ups.

Phase 1 uses encrypted unicast ESP-NOW for the short-range remote. Match Wi-Fi channel with the bike access point; use per-device keys and application-level acknowledgements. Phase 2 adds E220 UART radio after frequency, bandwidth, transmit power, antenna gain and airtime limits are validated for the UK. E220 is a radio transceiver, not merely an antenna; its antenna is distinct from GPS and ESP Wi-Fi antennas. No RF transmission is enabled in this starter.

A captive portal is a convenience page, not an authentication scheme or reliable presence detector. Phones sleep, switch networks and suspend web tabs. Use a paired application heartbeat with explicit PARK arming; link loss is advisory while riding. E220 is not an internet or cellular tracker: remote viewing needs a reachable paired receiver or a later network uplink.

3. Power and interference control

Never put the labelled 8.4 V battery output directly into Pico VSYS, ESP GPIO, the screen, GPS or a 5 V solenoid. Use fused, reverse-polarity-protected, appropriately rated regulated rails. The box's 5 V/2.1 A label is not evidence it can supply simultaneous lamp, radio and solenoid peak demand. Measure each load and regulator dropout/ripple; add peak currents and margin before choosing supply. Confirm matching protected cells, charger and battery protection separately.

Partition power into logic and load branches. Return solenoid, siren and lamp currents directly to the distribution point rather than through MCU ground wires. Keep low-level sensor wiring away from switching load loops and antennas. Twist reed signal with ground. Start with an external 4.7–10 kOhm pull-up to 3.3 V, a series resistor and RC/Schmitt conditioning; values must be checked against pulse width, cable capacitance and ESD requirements. A possible bench starting point is 1 kOhm series and 10 nF to ground at the input, then a 3.3 V Schmitt buffer. This is not a verified schematic. Never clamp a high-energy surge into GPIO rails.

Place 100 nF local bypass capacitors near logic supply pins plus suitable local bulk capacitance at radio and driver branches. Follow each regulator's stability requirements; do not add arbitrary capacitors across unknown switch-mode outputs. Use a logic-level MOSFET specified at the actual 3.3 V gate voltage, gate resistor and gate pull-down for each suitable DC load. GPIO must never power a coil or high-current LED. Put a suitably rated flyback diode at the solenoid, cathode to positive; validate release time, since a diode slows coil release. If release time requires a TVS/clamp, select it with the MOSFET voltage and coil energy in mind. Retain or replace the headlamp's proper constant-current driver and heatsink. Do not assume PWM on its supply is supported. A multi-function lamp may need isolated button emulation rather than directly switching individual LEDs.

Power loads through secured rated terminals, not solderless breadboard contacts. The breadboard is for logic and resistor-limited indicator LEDs. Start from a current-limited bench supply. Later use strain relief, waterproof connectors, sealed enclosures with appropriate condensation management, and secure mounting.

Interference acceptance test: count false wheel pulses with wheel stationary while cycling each load and transmitting radio; observe 3.3/5 V rails on an oscilloscope; log resets, rejected pulses, packet errors and GPS fix loss. Repeat with realistic cable lengths, low battery and simultaneous loads. Resolve electrical noise before loosening software filters. A software filter cannot prevent brownouts.

Earlier repository reviewed: yudlab/speedo, default beta-trial. Its speedo.ino has a 300 ms debounce interval, updates tLastRev before rejecting a pulse, and delays the main loop by 1100 ms. Those observations motivate a non-blocking design and separate accepted/rejected pulse counters; they do not establish the cause of past interference. At 2.1 m/revolution, 300 ms limits accepted pulses to roughly 25.2 km/h for one magnet.

4. Parking lock and alarm

Never engage a mechanism in a wheel, brake or steering system because a remote has disconnected. A broken reed cable looks like zero speed. GPS zero is also insufficient. Design a mechanical parking interlock that physically prevents engagement while riding, a separate PARK input, independent stillness evidence, position feedback and a manual release. No software-only guarantee is adequate.

Proposed sequence: explicit parking arm; healthy independent interlocks; loss of previously authenticated heartbeat for 5 seconds; 10-second visible/audible warning; recheck all conditions; bounded actuator request. Any movement, sensor fault, cancelled PARK, reset or returning heartbeat cancels pending actuation. Never infer lock position from a command. Show LOCKED only from verified feedback. No pulse duration is chosen until coil current/duty rating and mechanics are known. The code stops at LOCK_REQUEST_ONLY and has no solenoid driver or bypass flag.

Alarm has explicit arm/disarm and a bounded two-second visual bench test. Physical siren drivers, motion sensor selection and production alarm escalation are pending. Use a non-blocking scheduler; keep brake sensing alive during alarms and warnings.

5. Breadboard wiring proposal

Pico GPIO numbers, provisional allocation (not physical connector pin numbers):

Function	GPIO	Note
UART0 TX / RX	GP0 / GP1	Cross to ESP RX/TX, common ground, 3.3 V only
Reed contact	GP2	Dry contact to GND with conditioned input
Brake input	GP3	Dry test switch, independent of mechanical brake
Left / right	GP4 / GP5	Local maintained test switches
Physical PARK input	GP6	Reserved; not connected in starter
SPI1 SCK / MOSI	GP10 / GP11	Screen SCL means SPI clock; SDA means MOSI
TFT CS / DC / RST	GP9 / GP8 / GP12	Verify module labels and logic ratings
TFT backlight	GP13	Driver/control only after BLK circuit verified
Main / front idle / rear idle	GP14 / GP15 / GP16	Future external driver inputs
Left / right indicator groups	GP17 / GP18	Future front/rear driver grouping
Brake lamp / siren	GP19 / GP20	Future external drivers
Lock request / feedback	GP21 / GP22	Reserved, no actuator code
Independent stillness	Unassigned	Sensor selection and bus still pending

Test screen separately before combined harness. Do not infer ST7789 from size or GMT020-02 silkscreen alone. Confirm controller, dimensions, rotation, colour order, voltage and backlight current. The driver adapter is deliberately unbound. ESP pins are intentionally not guessed: obtain the exact S3-CAM schematic first.

6. Data, debug mode and implementation boundaries

Wheel formula: speed km/h = circumference_mm / magnets_per_rev * 3.6 / period_ms. Store distance in metric units regardless of display units. Measure tyre rollout under rider load, over multiple revolutions; average and enter millimetres. Ignore implausibly close pulses without moving the accepted timestamp. No pulses causes a display timeout, not a safety assertion of standstill.

Wired prototype protocol: newline JSON at 115200 baud, version, sequence, monotonic ms, kmh, trip_mm, rejected count and output state. Proposed production transport adds bounded framing, CRC, resynchronisation, sequence checks, boot/session identity, heartbeats and explicit acknowledgements. CRC detects corruption, not attackers. Wireless commands require authenticated encryption, replay prevention, paired identity and expiry; never accept arbitrary pin numbers or raw actuator writes.

Persistent records belong on the ESP: schema version, trip ID, timestamps with time validity, distance, duration, max speed, fix age/quality and optional positions. Use batched writes with checksums and two-slot recovery or a journal; avoid writing flash per wheel pulse. GPS logging should be opt-in, bounded and exportable; never commit tracks or keys. The starter does not yet parse NMEA or persist ride statistics.

Production debug mode should require a physical service jumper, stationary parked bike, authenticated local session and a short timeout. Individual lamp/siren tests must be bounded; test-all must sequence channels rather than energise every load at once. Lock interlocks remain mandatory. The supplied web app is simulation only, with fault injection and JSON session export; no endpoint can drive GPIO.

7. Build stages and acceptance

1. Restore brakes and investigate chain wear, indexing, stiff links, cassette wear and freehub symptoms. Check housing and cable compatibility before fitting.

2. On USB power, prove Pico reed sensing and calibration. Verify 100 wheel turns match expected distance, bounce is rejected and stopped display times out.

3. Identify TFT and integrate its driver. Stress refresh while counting pulses.

4. Inspect lamp assembly; measure voltage/current and establish isolated controls. Test each external driver with a dummy load first; then actual lamps off-bike.

5. Integrate UART framing, GPS parser and durable records; test malformed frames, stale fixes, MCU resets and power loss mid-save. Prove local lights survive ESP loss.

6. Pair encrypted short-range remote, screen and trackball. Test acknowledgements, retries, duplicate commands, wrong peers and phone sleep behaviour.

7. Validate E220 UK parameters and antenna matching before RF tests. Measure airtime including retries and acknowledgements; rate-limit accordingly.

8. Prototype parking mechanics away from the bicycle. Prove independent interlocks, manual release and position feedback. No moving-wheel actuation tests with a rider.

9. Replace breadboard with soldered secured construction; repeat interference, vibration and weather tests before considering road use.

8. Sources and open decisions

Manufacturer / regulator references, checked 10 November 2026:

- Raspberry Pi Pico documentation: https://www.raspberrypi.com/documentation/microcontrollers/pico-series.html

- Espressif ESP32-S3 datasheet: https://documentation.espressif.com/esp32_s3_datasheet_en.pdf

- ESP-NOW reference: https://docs.espressif.com/projects/esp-idf/en/stable/esp32s3/api-reference/network/esp_now.html

- Ebyte E220-900T22D manual: https://www.ebyte.com/Uploadfiles/Files/2021-7-6/202176188184448.pdf

- Ofcom short-range devices: https://www.ofcom.org.uk/spectrum/radio-equipment/short-range-devices

- Ofcom IR2030: https://www.ofcom.org.uk/siteassets/resources/documents/spectrum/interface-requirements/ir-2030.pdf

- Earlier project: https://github.com/yudlab/speedo/tree/beta-trial

UK radio compliance depends on the exact permitted sub-band, emission bandwidth, antenna-adjusted radiated power and channel-access/duty-cycle rules. Do not assume factory 900/915 MHz or full 22 dBm settings are allowed. Exact E220 settings remain a design gate; no blanket legality claim is made here. Road lighting compliance also needs review against the final lamps and installation, not the concept image.

Need before hardware integration: exact TFT driver; ESP board schematic; lamp supply and pinout; solenoid rating and mechanism; remote screen and second ESP availability; battery cell/protection details; measured wheel circumference. These unknowns do not prevent the simulation or breadboard input demo from being used now.